Harsh Jain

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Summary

Applied Machine Learning and AI student with strong experience building end-to-end ML systems, including data pipelines, model training, optimization, and deployment using FastAPI and Docker. Hands-on with real-world projects in adversarial RL, LLM fine-tuning, manufacturing traceability, and emissions prediction, with a focus on scalable APIs, clean system design, and measurable performance improvements.

Education

Savitribai Phule Pune University (SPPU) Aug 2022 – May 2027

B.E. Artificial Intelligence & Data Science | **GPA: 8.75/10**

Technical Skills

Languages & APIs: Python, SQL, JavaScript/TypeScript, FastAPI, Next.js 15, React **ML / RL:** PyTorch, GRPO, PPO, LoRA/QLoRA, TRL, Unsloth, Ray RLlib, HuggingFace Transformers, PEFT

Computer Vision: YOLOv8, ONNX Runtime, OpenCV, Albumentations **Infra / DB:** Docker, GitHub Actions, Google Cloud, PostgreSQL, MongoDB, Redis

Other: RAG Pipelines, RLHF, Adversarial RL, Agentic AI

Projects

ConflictBench | GRPO · RL · Qwen2.5-3B · LoRA · Unsloth · TRL · Gradio

- Built an **RL environment** that trains language models to resolve contradictory business instructions by learning an implicit **6-tier authority hierarchy** (Legal > C-Suite > VP > Director > Team Lead > IC) entirely from a reward signal across episodes of **8–28 directives** with **2–6 embedded conflict pairs** — the hierarchy is never stated in the prompt.
- Designed a **5-rubric deterministic reward function** (no LLM judge) covering final-state correctness, contradiction-freedom, conflict-pair identification F1, efficiency, and JSON schema compliance; fine-tuned **Qwen2.5-3B** with GRPO + LoRA (r=32) on 400 scenarios over 2 epochs on an A100 48 GB, lifting composite reward from **0.14** → **0.50 (+257%)** over the zero-shot baseline with no prior SOTA to compare against.
- Published LoRA adapter and browser-runnable base-vs-fine-tuned demo on HuggingFace. **Finalist** — Meta × PyTorch × HuggingFace OpenEnv Hackathon, Bangalore; **Top 100** — Scaler School of Technology pre-selection round.

ARMS RACE — Adversarial Oversight Arena | PPO · LoRA · PEFT · REINFORCE · SFT

- Built a **two-agent zero-sum adversarial loop** where a Red Agent generates semantically plausible silent-failure hallucinations and a Blue Agent acts as a **factual gatekeeper**, producing structured JSON verdicts (Pass / Flag / Probe) to decide whether a model response should be trusted.
- Designed **Expert Correction Training (ECT)**: on Blue Agent failure, a programmatically generated "perfect" JSON response is injected as a supervised update, converting failed RL steps into stable learning signal and preventing policy collapse — hallucination detection improved from **25%** → **100%** with a **4% false-alarm rate** and **96% OOD generalisation**.
- Structured **asymmetric rewards** (TP +0.6, FP −2.0, FN −0.6) to make trivially conservative strategies like flagging everything mathematically unprofitable; used zero-sum ELO tracking to prevent co-declining agents.

TraceLink | FastAPI · React · Firebase Auth · SQLite · Docker

- Built a **production-deployed manufacturing traceability system** that lets quality teams trace any dispatch order backward through production batches, QC inspections, and raw material lots across **6 entity types** — and forward from a flagged raw lot to all potentially affected customer orders for rapid recall investigation.
- Implemented **6-role RBAC** (operator, supervisor, quality, manager, admin) with Firebase ID-token verification, CSV ingestion with **full rollback support**, a request-level audit trail for every action, and a **natural-language AI query endpoint** for non-technical users.
- Containerised into a **single Docker image** (Bun frontend + FastAPI backend) and deployed on Render with auto-deploy from the main branch; OpenAPI docs available at `/api/docs`.

Arivon — Adaptive Learning Platform | Next.js 15 · FastAPI · Groq Whisper · Haystack RAG · MongoDB · Redis

- Built an adaptive learning platform that detects **metacognitive miscalibration** — when a student's self-assessed confidence diverges from their actual performance — and dynamically adjusts the learning path, question difficulty (via **Bloom's taxonomy engine**), and study recommendations.
- Integrated a **voice-based exam interface** using Groq Whisper, a RAG-powered study mentor via Haystack, and a **React Flow knowledge graph** mapping concept relationships; backed by a **PostgreSQL**/**MongoDB**/**Redis** stack for session state, embeddings, and low-latency caching.
- **3rd Place** at Pragyantra, PES Modern College of Engineering in Domain.

SO_2 Emission Prediction System | XGBoost · FastAPI · Docker · PostgreSQL · Optuna 2024

- Architected an **end-to-end ML pipeline** predicting SO_2 emissions from Indian coal power plants, achieving 85% accuracy on held-out test data via cross-validation and feature selection.
- Applied **Optuna-based hyperparameter tuning** on XGBoost and deployed the model as a containerised **FastAPI microservice** on Docker for reproducible, scalable cloud deployment.
- Boosted prediction efficiency by 20% through **feature engineering and pipeline automation**; authored comprehensive REST API documentation with structured error handling and input validation.

Hackathons & Awards

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Top 100 — Scaler School of Technology OpenEnv Pre-Selection | ConflictBench

3rd Place — Pragyantra, PES Modern College of Engineering | Arivon

Certifications

Deep Learning Specialization — Andrew Ng / deeplearning.ai

100 Days Of Code:The Complete Python Pro Bootcamp — Udemy

Generative AI with LLMs — AWS / Coursera

LLM Fundamentals — Hugging Face